Exactness and minimal architectures for differentiable sliced optimal transport plans

Optimal transport (OT) matches two distributions through a transport *plan*; recent "sliced" methods approximate this plan cheaply by optimizing one-dimensional projections. When such an approximate plan *exactly* equals the true OT plan and how expressive a projection this requires is not understood. This internship characterizes exact recovery and the smallest projection architecture that achieves it, validates the predictions numerically, and opens the harder quantitative question of the error when recovery is only approximate.

§1. Organization

- **Start date**: As soon as possible (flexible).
- **Localization**: IRISA, Rennes.
- **Supervision**:
  - Romain Tavenard (Prof., Université Rennes 2, MALT Inria team, IRISA, Rennes).
  - Laetitia Chapel (Prof., Institut Agro, TAGADA research group, IRISA, Vannes).
  - Paul Viallard (Researcher, MALT Inria team, IRISA, Rennes).

§2. Context

Optimal transport (OT) compares two probability distributions by computing the cheapest way to move the mass of one onto the other. It returns two things at once: a number, the *distance* (how costly the move is), and a *plan* (which mass goes where). The plan is what makes OT so useful in practice for matching, alignment, or generative modelling but computing it exactly becomes very expensive as the dimension grows.

Sliced OT is a popular way around this cost. The idea is to project the two distributions onto a one-dimensional line, where OT has a cheap, closed-form solution: one simply sorts the projected points, and the resulting plan is a permutation matching them in order. To turn this one-dimensional plan into a plan in the original space, one *lifts* it back, transporting the original points according to the matching found on the line. DGSWP (Chapel et al., 2025; Mahey et al., 2023), does exactly this and, rather than picking the projection at random, *learns* the (possibly non-linear) *projector* that makes the lifted plan best approximate the true OT plan, casting the search as a differentiable bilevel problem.

These developments are largely *algorithmic*. The supporting theory, where it exists, concerns the sliced *distance* and says nothing about the *plan*. The most basic question for the plan is still open: when does the lifted, projector-optimized plan *exactly* equal the true OT plan, and how expressive must the projector be for this? Answering it says whether the method is principled or merely a heuristic, tells a practitioner how rich a projector to use, and is the first step toward bounding the gap when exactness fails.

The core object is the map from a projector to the lifted plan it induces. The central question is constructive: **for which data, and with how expressive a projector, does this plan exactly equal an optimal plan?**

This pairs an exactness question with a capacity one. Establishing exactness fixes *which* plans a class of projectors can realize; the matching question is *how rich* a projector this takes, in terms of minimal width, depth, or number of projections. This is a representation/capacity statement in the spirit of neural-network memorization (Park et al., 2021) and approximation theory (Yarotsky, 2017), but for transport plans rather than function values, and it is open.

Beyond exactness lies the harder, quantitative question of how the error behaves when the plan is only *approximately* OT, with neural-network approximation theory (Yarotsky, 2017) as the natural starting point and the optimization-landscape question below as its companion.

§3. Goals

Concretely, the candidate will:

1. formalize the lifted-plan map and a precise notion of exactness that accounts for non-uniqueness of the optimal plan;
2. turn this into a minimal-capacity statement, i.e. find the smallest projector (width, depth) that realizes the OT plan while remaining a valid sliced geometry, drawing on neural-network capacity and approximation results (Park et al., 2021; Yarotsky, 2017);
3. validate the predicted conditions and architectures with controlled experiments on non-trivial geometry, measuring the lifted-plan-to-OT gap;
4. (open direction / Ph.D.) move beyond exact recovery toward quantifying the approximation error, and study the optimization landscape of the projector search so that the solver provably *reaches* a good projector.

Candidates should possess:

- a strong mathematical background (analysis, linear algebra, convexity; notions of optimal transport are a plus),
- the ability to formulate and prove characterization results,
- good Python coding skills to validate the theory numerically,
- ability to communicate in English in a scientific context.

Funding for a Ph.D. has already been secured: upon a successful internship, the candidate will be offered the opportunity to continue this work as a fully funded doctoral thesis. A successful candidate will develop or reinforce skills key to that continuation, namely:

- a working command of the geometry of optimal transport and sliced approximations,
- experience proving theoretical results and stating them rigorously,
- the habit of pairing theoretical conditions with numerical validation.

§4. Bibliography

Bonneel, N., Rabin, J., Peyré, G., & Pfister, H. (2015). *Sliced and Radon Wasserstein Barycenters of Measures.* Journal of Mathematical Imaging and Vision.

Chapel, L., Tavenard, R., & Vaiter, S. (2025). *Differentiable Generalized Sliced Wasserstein Plans.* NeurIPS. arXiv:2505.22049.

Kolouri, S., Nadjahi, K., Şimşekli, U., Badeau, R., & Rohde, G. K. (2019). *Generalized Sliced Wasserstein Distances.* NeurIPS.

Mahey, G., Chapel, L., Gasso, G., Bonet, C., & Courty, N. (2023). *Fast Optimal Transport through Sliced Wasserstein Generalized Geodesics.* NeurIPS.

Park, S., Lee, J., Yun, C., & Shin, J. (2021). *Provable Memorization via Deep Neural Networks using Sub-linear Parameters.* Proceedings of the 34th Annual Conference on Learning Theory (PMLR).

Yarotsky, D. (2017). *Error Bounds for Approximations with Deep ReLU Networks.* Neural Networks.